Linear Equations & Slope Error Analysis

TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Slope formula

ORIGINAL PROBLEM

Find the slope through (2, 3) and (6, 11).

INCORRECT WORKED SOLUTION

$$m = \frac{\text{change in x}}{\text{change in y}}$$

$$m = \frac{6 - 2}{11 - 3}$$

$$m = \frac{4}{8} = \frac{1}{2}$$

A. EXPLAIN THE MISTAKE (1 POINT)

Slope is rise over run: $\frac{\text{change in y}}{\text{change in x}}$. The wrong work put x over y.

B. SHOW CORRECT WORK (1 POINT)

$$m = \frac{\text{change in y}}{\text{change in x}}$$

$$m = \frac{11 - 3}{6 - 2}$$

$$m = \frac{8}{4} = 2$$

2. Order of subtraction

ORIGINAL PROBLEM

Find the slope through (1, 5) and (4, −1).

INCORRECT WORKED SOLUTION

$$m = \frac{y_2 - y_1}{x_1 - x_2}$$

$$m = \frac{-1 - 5}{1 - 4}$$

$$m = \frac{-6}{-3} = 2$$

A. EXPLAIN THE MISTAKE (1 POINT)

The points must be subtracted in the SAME order on top and bottom. The bottom was subtracted in the opposite order, which flipped the sign.

B. SHOW CORRECT WORK (1 POINT)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{-1 - 5}{4 - 1}$$

$$m = \frac{-6}{3} = -2$$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Reading slope-intercept form

ORIGINAL PROBLEM

Identify the slope and y-intercept of $y = 5 - 2x$.

INCORRECT WORKED SOLUTION

$y = 5 - 2x$

$y = mx + b$

The first number is the slope: $m = 5$

The second number is the y-intercept: $b = -2$

A. EXPLAIN THE MISTAKE (1 POINT)

In $y = mx + b$, the slope is the coefficient of x and b is the constant. Rewrite as $y = -2x + 5$: $m = -2$, $b = 5$.

B. SHOW CORRECT WORK (1 POINT)

$y = 5 - 2x$

$y = -2x + 5$

$m = -2$

$b = 5$ (the point (0, 5))

4. Solving for y

ORIGINAL PROBLEM

Write $3x + 2y = 12$ in slope-intercept form.

INCORRECT WORKED SOLUTION

$3x + 2y = 12$

$2y = 12 + 3x$

$y = 6 + \frac{3x}{2}$

$y = \frac{3}{2}x + 6$

A. EXPLAIN THE MISTAKE (1 POINT)

To remove 3x from the left side, SUBTRACT 3x from both sides: $2y = 12 - 3x$. The sign of the x-term was lost.

B. SHOW CORRECT WORK (1 POINT)

$3x + 2y = 12$

$2y = -3x + 12$

$y = -\frac{3}{2}x + 6$

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5. Writing an equation

ORIGINAL PROBLEM

Write the equation of the line with slope 3 through (2, 5).

INCORRECT WORKED SOLUTION

$y = mx + b$

$y = 3x + b$

The line passes through (2, 5), so b = 5.

$y = 3x + 5$

A. EXPLAIN THE MISTAKE (1 POINT)

The y-coordinate of a point is NOT automatically the y-intercept. Substitute the point to find b: $5 = 3(2) + b$.

B. SHOW CORRECT WORK (1 POINT)

$y = 3x + b$

$5 = 3(2) + b$

$5 = 6 + b$

$b = -1$

$y = 3x - 1$

6. Parallel lines

ORIGINAL PROBLEM

Write the line parallel to $y = 2x + 1$ through (4, 3).

INCORRECT WORKED SOLUTION

Parallel lines have opposite reciprocal slopes.

$m = -\frac{1}{2}$

$3 = -\frac{1}{2}(4) + b \rightarrow b = 5$

$y = -\frac{1}{2}x + 5$

A. EXPLAIN THE MISTAKE (1 POINT)

Parallel lines have EQUAL slopes. Opposite reciprocal slopes belong to perpendicular lines. The slope should be 2.

B. SHOW CORRECT WORK (1 POINT)

Parallel lines have equal slopes: $m = 2$

$3 = 2(4) + b$

$b = -5$

$y = 2x - 5$

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7. Perpendicular lines

ORIGINAL PROBLEM

Find the slope of a line perpendicular to $y = \frac{2}{3}x - 4$.

INCORRECT WORKED SOLUTION

The slope of the given line is $\frac{2}{3}$.

Perpendicular slopes are opposites.

$m = -\frac{2}{3}$

A. EXPLAIN THE MISTAKE (1 POINT)

Perpendicular slopes are OPPOSITE RECIPROCALS: flip the fraction AND change the sign. Check: $\frac{2}{3} \cdot (-\frac{3}{2}) = -1$.

B. SHOW CORRECT WORK (1 POINT)

The slope of the given line is $\frac{2}{3}$.

Flip the fraction and change the sign.

$m = -\frac{3}{2}$

Check: $\frac{2}{3} \cdot (-\frac{3}{2}) = -1$

8. Finding an intercept

ORIGINAL PROBLEM

Find the x-intercept of $y = 2x - 6$.

INCORRECT WORKED SOLUTION

$y = 2x - 6$

Let $x = 0$: $y = 2(0) - 6 = -6$

The x-intercept is -6.

A. EXPLAIN THE MISTAKE (1 POINT)

The x-intercept is where the line crosses the x-axis, so $y = 0$ (not $x = 0$). Set $0 = 2x - 6$.

B. SHOW CORRECT WORK (1 POINT)

$y = 2x - 6$

Let $y = 0$: $0 = 2x - 6$

$6 = 2x$

$x = 3$

The x-intercept is (3, 0).

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9. Vertical lines

ORIGINAL PROBLEM

Find the slope of the line through (3, 2) and (3, 7).

INCORRECT WORKED SOLUTION

$$m = \frac{7 - 2}{3 - 3}$$

$$m = \frac{5}{0}$$

$$m = 0$$

A. EXPLAIN THE MISTAKE (1 POINT)

Dividing by 0 is undefined, and a vertical line has UNDEFINED slope. A slope of 0 is a horizontal line.

B. SHOW CORRECT WORK (1 POINT)

$$m = \frac{7 - 2}{3 - 3}$$

$$m = \frac{5}{0}$$

Division by zero is undefined.

The line $x = 3$ is vertical: slope is undefined.

10. Point-slope form

ORIGINAL PROBLEM

Write the equation of the line with slope 2 through $(-3, 4)$ in point-slope form.

INCORRECT WORKED SOLUTION

$$y - y_1 = m(x - x_1)$$

$$y - 4 = 2(x - 3)$$

A. EXPLAIN THE MISTAKE (1 POINT)

The point has $x_1 = -3$, so $x - x_1 = x - (-3) = x + 3$. The sign of the x-coordinate was dropped.

B. SHOW CORRECT WORK (1 POINT)

$$y - y_1 = m(x - x_1)$$

$$y - 4 = 2(x - (-3))$$

$$y - 4 = 2(x + 3)$$

Check: at $x = -3$, $y = 4$

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